

WEN-YI (LEITH) LO

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EDUCATION

NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY (NTUST)

Taipei, Taiwan

B.S. in Computer Science and Information Engineering

Jul 2020 – Jun 2023

GPA: 4.01/4.3 (Top 15%)

Honors: Academic Excellence Award, 2020, 2021 (Top 3 out of 40+ students)

Related Courses: Algorithms (95), Data Structures (95), Operating Systems (95), Database Systems (95), Computer Networks (95), Software Engineering (87)

B.S. in Mechanical Engineering

Sep 2019 – Jun 2020

PROFESSIONAL EXPERIENCE

GARMIN ASIA – A global leader in GPS technology and wearable devices (NASDAQ: GRMN)

Linkou, Taiwan

Software Engineer Intern, Platform Software Department

Jul 2024 – Jun 2025

- Developed an internal Web Security Management system (Laravel, Vue, MySQL) that automated quarterly audit workflows, cutting manual permission cross-checking time by at least 50%
- Led front-end design and feature decomposition in Phase 3 of the Web Security Management system while contributing to full-stack development, ensuring timely delivery of project milestones
- Prototyped an MQTT-based vehicle communication system (Django, Bootstrap, AWS IoT, Android) by building the web interface and message pipeline, validating the feasibility of replacing legacy hardware MQTT devices
- Individually engineered an automated Timecard Reminder System (Python, Jenkins) for the 320-person Auto OEM division, reducing monthly manual notifications from 60 to <10 cases and streamlining reporting for managers
- Presented the MQTT system at the final internship showcase to an audience of 20+ HR representatives, supervisors, and interns, receiving a premium offer that allowed direct interview access by bypassing standard resume screening

INDUSTRIAL TECHNOLOGY RESEARCH INSTITUTE – Taiwan's leading research institute

Hsinchu, Taiwan

Research Assistant, Software Design Department

Jul 2024 – Jan 2025

Project: Congestion Control in Satellite Network

- Delivered 14 bi-weekly presentations on congestion control in satellite networks, synthesizing research papers into accessible insights for supervisors, professors, and engineers
- Analyzed algorithmic differences and parameters (e.g., Slow Start, Cubic) and provided comparative evaluations, guiding supervisors to better understand trade-offs in past congestion control approaches

INDUSTRY-ACADEMIA COLLABORATION

NTUST & IGS (INTERNATIONAL GAMES SYSTEM) – Leading arcade and gaming solution provider in Asia

Project: Track Editor Migration (Unity to Unreal Engine 5)

Sep 2022 – Mar 2023

- Migrated a Unity-based track-wall generation module from C# to C++ in Unreal Engine 5, ensuring functional equivalence and smooth transition during engine migration
- Enhanced the track editor by adding user-defined wall height/width adjustments, implementing undo/redo operations, and introducing magnetic snapping, which improved usability and maintained a stable 60fps rendering performance
- Collaborated in a 5-member student team to deliver the complete source code and final technical report to the industry partner, achieving project acceptance and approval from the partner's PM team

TEACHING EXPERIENCE

NATIONAL TAIWAN UNIVERSITY OF SCIENCE AND TECHNOLOGY (NTUST)

Taipei, Taiwan

Teaching Assistant, Object-Oriented Programming & Lab (Undergraduate Courses)

Jan 2023 – Jun 2023

- Supported instruction for 130 undergraduates in C++ programming by preparing lab materials and clarifying course concepts, improving student comprehension of object-oriented design principles
- Evaluated weekly assignments, midterm and final exams, and project submissions through automated testing systems, ensuring fair grading and timely feedback
- Led project evaluation of a Chess game and co-graded Minesweeper and Pokémon projects, strengthening student ability in applied algorithm design and code quality review

Teaching Assistant, Introduction to Next-Generation Wireless Networks (Undergraduate)

Sep 2023 – Dec 2023

- Assisted in teaching 70 undergraduates networking fundamentals through Python and C++ assignments, providing feedback that reinforced practical protocol implementation skills
- Designed and graded course projects, including a Wireshark packet analysis lab and a Python socket programming assignment, enabling students to gain hands-on experience in networking and systems programming
- Provided consistent assignment evaluations and guidance, contributing to a supportive learning environment and improved student performance across the course

PROJECTS

ROAD EDITOR – NTUST

Taipei, Taiwan

Undergraduate Capstone Project

Jan 2022 – Jan 2023

- Designed and implemented a 2D road editor with puzzle-style assembly of road segments and procedural generation of roadside objects (trees, buildings)
- Integrated user controls for translation, rotation, and scaling, increasing flexibility in road design and user experience
- Presented at the NTUST Undergraduate Exhibition, showcasing technical execution and collaborative teamwork

KOTLIN SUBSET COMPILER – NTUST

Taipei, Taiwan

Compiler Design Course Project

Jun 2022

- Developed a lexer in C using flex with a custom symbol table to tokenize Kotlin source code, enabling grammar parsing and AST generation
- Implemented a parser in yacc/bison to process arithmetic expressions and control structures, improving understanding of compiler design and parsing theory
- Generated JVM bytecode from the AST, allowing execution of compiled Kotlin subset programs directly on the JVM

AMUSEMENT PARK – NTUST

Taipei, Taiwan

Introduction to Computer Graphics Course Project, C++, OpenGL, GLSL

Jan 2022

- Created a 3D amusement park with animated rides (roller coaster, Ferris wheel, coffee cups) and multi-camera perspectives, providing immersive simulation features
- Implemented model loading and visual effects such as fireworks animation, enhancing realism and interactivity
- Achieved Honorable Mention for project excellence, ranking among the top 5 of 12 teams

HONORS & LEADERSHIP EXPERIENCE

- **Academic Excellence Scholarship, NTUST (2023–2024):** Merit-based scholarship awarded to the top 5% of students for outstanding academic performance and potential for future studies
- **PDAO Programming Competition, National Taiwan University (2021):** Placed 8th among 68 teams in an algorithmic programming contest comparable to LeetCode challenges, demonstrating problem-solving and coding efficiency under time constraints
- **Leadership:** NTUST Kyudo Club – Finance Officer & Team Member, managed the club's annual budget and resource allocation while representing the team in the freshman tournament, securing 3rd place; strengthened leadership, discipline, and team collaboration through competitive training and performance